

Applied Mathematics for Scientists

Science

May, 2026

Applied Mathematics for Scientists (A11530)

Short Title: Applied Maths for Scientists
Faculty Science and Computing
Department Science
Cluster Mathematics and Physics
Author FLEONARD
Credits 5

Level: Introductory

Description of Module / Aims

This module looks at modelling a range of scientific phenomena using differential equations such as population growth, bacterial destruction, heat transfer, drug diffusion, viral propagation, radioactive decay, kinematics, etc. An introduction to partial differentiation is provided, with applications to optimisation. Multiple integrals are presented and are applied to the evaluation of volumes.

Programmes

MTHE-0044	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	2 / 4 / M
MTHE-0044	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	2 / 4 / M
MTHE-0044	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	2 / 4 / M
MTHE-0044	BSc in Pharmaceutical Science (WD_SPHSC_D)	2 / 4 / M

Indicative Content

- Calculus of several variables (differential and integral): find and classify stationary points on a surface; applied optimisation problems; evaluate multiple integrals
- Differential equations: modelling dynamical systems; developing differential equations; various methods for solving differential equations
- Plots: plotting the solution to a differential equation

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Calculate and apply partial derivatives to optimisation problems.
2. Calculate multiple integrals.
3. Employ various techniques to solve differential equations.
4. Construct a plot of a solution to a differential equation.

Learning and Teaching Methods

- Lectures.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	1, 2, 3, 4
<i>In-Class Assessment</i>	80%	1, 2, 3
<i>Assignment</i>	20%	4

Assessment Criteria

- <40%:** Unable to do multiple variable calculus or manipulate differential equations.
- 40%-49%:** Able to do basic multiple variable calculus and manipulate simple differential equations.
- 50%-59%:** Able to solve a variety of multiple variable calculus problems and manipulate a range of differential equations.
- 60%-69%:** Able to tackle a comprehensive range of theoretical and applied problems in multiple variable calculus and in differential equations.
- 70%-100%:** As above and have an authoritative understanding of the methods and applications of multiple variable calculus and in differential equations.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Independent Learning	99	

Essential Material

- "Moodle." moodle.wit.ie

Supplementary Material

- "Wolfram Mathworld." <http://mathworld.wolfram.com/>
- Bronson, R. _Differential Equations, _Schaum's__easy outlines_. 4. USA: McGraw-Hill, 2014.
- Stroud, K.A. _Engineering Mathematics_. 6. China: Palgrave Macmillan, 2007.